Fpt University

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# CAPSTONE PROJECT REPORT

## Report 5 – Software Test Documentation

– FPT University, October 2026 –

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# I. Record of Changes

| Date | A* M, D | In charge | Change Description |
|---|---|---|---|
| 26/10/2026 | M | Lê Nguyễn Quang Đức | Populate VISTA Review 2 content on original university template; code verification pending |
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*A - Added M - Modified D - Deleted

# II. Testing Documentation

## 1. Scope of Testing

Test Scope: authorization, learning administration, assignments, program versioning, schematic experiment and symbolic evaluation, feedback/reflection. Physical robot end-to-end and staging deployment are pending validation; no fabricated outcomes.

Test Levels: unit (Jest/Vitest/Python), integration (NestJS service and Prisma/migrations), system/acceptance (four workflow main and exception cases).

Assumptions: seeded isolated database, controlled role-based accounts, no real secrets in repository; robot runtime safety tests must not activate motion outside authorized lab.

## 2. Test Strategy

Use unit tests for guards/service decisions, integration tests with scoped seeded PostgreSQL fixtures, system tests for UI/API end-to-end flows, and acceptance demonstrations with owned scenarios. Physical robot test results require controlled hardware evidence and safety approval.

Tests cover main and exception paths plus negative cross-organization access, immutable version ownership, race/conflict and dependency unavailability.

### 2.1 Testing Types

Each test specifies feature/FR, actor, preconditions, steps, expected persisted state, negative result and automation target.

### 2.2 Test Levels

Unit checks, integration endpoints/transactions, manual staging e2e and fault/retry tests.

| Type of Tests | Test Level | | | |
|---|---|---|---|---|
| | Unit | Integration | System | Acceptance |
| <Test Type1> | X | X | X | X |
| <Test Type2> | | | X | X |
| <Test Type3> | X | X | | |

### 2.3 Supporting Tools

Supporting tools: GitHub Actions, Jest, Vitest, Python unittest/pytest where actually configured, Prisma validation, database fixtures.

| Purpose | Tool | Vendor/In-house | Version |
|---|---|---|---|
| Backend unit and integration | Jest | Open source | Version per lockfile |
| Frontend unit | Vitest or configured runner | Open source | Verify package.json |

# 3. Test Plan

## 3.1 Human Resources

Human Resources: Đức test ownership/integration; Đạt learning; Tuấn robot safety/runtime; Lợi assessment/data; independently collect logs.

| Worker/Doer | Role | Specific Responsibilities/Comments |
|---|---|---|
| Lê Nguyễn Quang Đức | Backend/ QA lead | Auth/scope and CI evidence |
| Trần Tiến Đạt | WF-02 tester | Assignment/submission tests |

## 3.2 Test Environment

Test environments: local React/Vite + NestJS + PostgreSQL/Supabase staging, optional isolated Python runtime. Do not use live production data for destructive tests.

| Purpose | Tool | Provider | Version |
|---|---|---|---|
| Development | React/ Vite + NestJS | Repository | Version in package.json |
| Database | Supabase PostgreSQL | Supabase | Verify live instance |

## 3.3 Test Milestones

Milestones: Iteration 1 CI/test demonstration before review-2 tag; Defense 1.1 full staging workflows according to BM04-approved targets.

| Milestone Task | Start Date | End Date |
|---|---|---|
| Iteration 1 QA | Before review-2 tag | Before review-2 tag |
| Defense 1.1 workflows | After Review 2 | Per BM04 approval |

# 4. Test Cases

WF-01 authorized allocation and cross-school denial; WF-02 classroom publishing and student submission; WF-03 schematic pending creation / invalid mode rejection; WF-04 teacher review and unauthorized assessment denial. Each must have preconditions, expected effects, assertions, and execution proof. Report5 spreadsheet holds detailed cases.

Use supplied original Report5_Test Report.xlsx and Report5_Unit Test.xls templates; do not add arbitrary sheets. Initial design cases are Pending until executed.

- *Unit Test Cases: Report5_Unit Test.xls*
- *Other Test Cases (IT, ST, AT): Report5_Test Report.xls]*

## 5. Test Reports

No verified executed test result supplied in this session. Preserve status Not Run/Pending and record actual run time, commit SHA, log, tester, defect and outcome only after execution.

Test results cannot be described as passed until test commands, run time, commit, logs and defect IDs are captured.

## Diagram consistency test cases — Review 2

• ERD-01: Verify all physical PK/FK types and referential constraints from current PostgreSQL migrations.

• ERD-02: Verify optionality and cardinality against nullable and unique constraints.

• STATE-01: Match all four State Machines to implemented enums and service transitions.

• WF-03-01: Education API rejects unsupported execution modes; no physical execution result is assumed.

• WF-01-01: Classroom robot allocation rejects foreign-school and non-entitled resources.